

# Prime Number Identifier Circuit

## PC/CP220 Lab 6

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### Description:

Finding prime numbers is a common task in mathematics. This circuit will identify the prime numbers between zero and fifteen. A number is defined as prime if it has exactly two divisors, itself and one.

## FROM LAB 4:

### Inputs:

```
t1: not(a3) and not(a2) and a1;
t2: not(a3) and a2 and a0;
t3: not(a2) and a1 and a0;
t4: a2 and not(a1) and a0;

prime: t1 or t2 or t3 or t4;

prime, a0=false, a1=false, a2=false, a3=false;

prime, a0=true, a1=false, a2=false, a3=false;
prime, a0=false, a1=true, a2=false, a3=false;
prime, a0=true, a1=true, a2=false, a3=false;
prime, a0=false, a1=false, a2=true, a3=false;
prime, a0=true, a1=false, a2=true, a3=false;

prime, a0=false, a1=true, a2=true, a3=false;
prime, a0=true, a1=true, a2=true, a3=false;
prime, a0=false, a1=false, a2=false, a3=true;
prime, a0=true, a1=false, a2=false, a3=true;
prime, a0=false, a1=true, a2=false, a3=true;

prime, a0=true, a1=true, a2=false, a3=true;
prime, a0=false, a1=false, a2=true, a3=true;
prime, a0=true, a1=false, a2=true, a3=true;
prime, a0=false, a1=true, a2=true, a3=true;
prime, a0=true, a1=true, a2=true, a3=true;
```

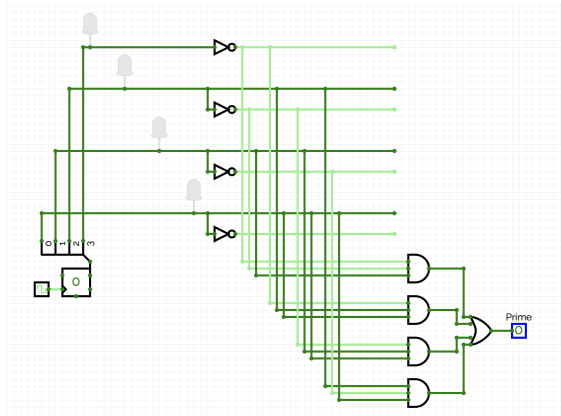
### Output:

```
 $\neg a_3 \wedge \neg a_2 \wedge a_1$ 
 $\neg a_3 \wedge a_2 \wedge a_0$ 
 $\neg a_2 \wedge a_1 \wedge a_0$ 
 $a_2 \wedge \neg a_1 \wedge a_0$ 
 $\neg a_3 \wedge \neg a_2 \wedge a_1 \vee \neg a_3 \wedge a_2 \wedge a_0 \vee \neg a_2 \wedge a_1 \wedge a_0 \vee a_2 \wedge \neg a_1 \wedge a_0$ 

false
false
true
true
false
true
false
true
false
false
true
false
false
true
false
true
false
false
```

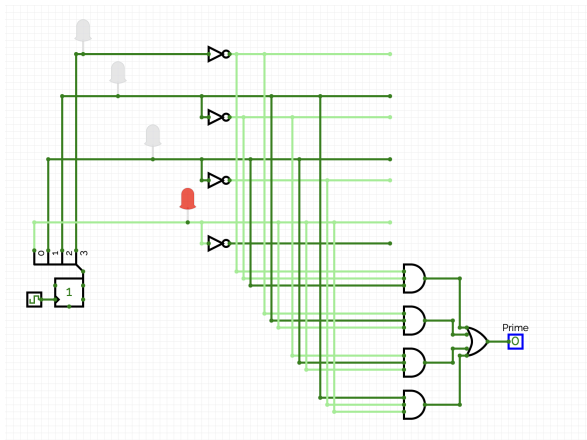
## Circuit & Explanation:

### Number 0:



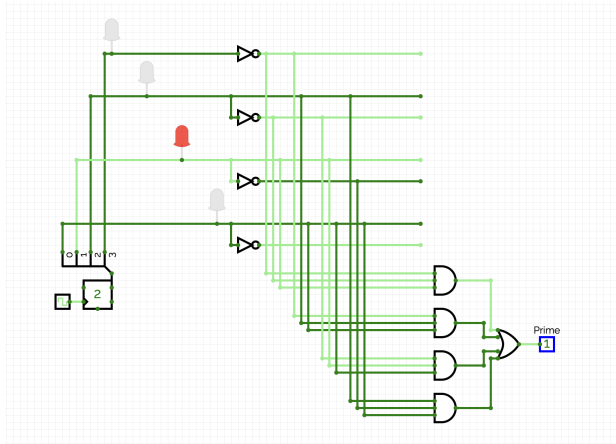
The Number 0 is neither prime nor composite which is why the output is False (0)

### Number 1:



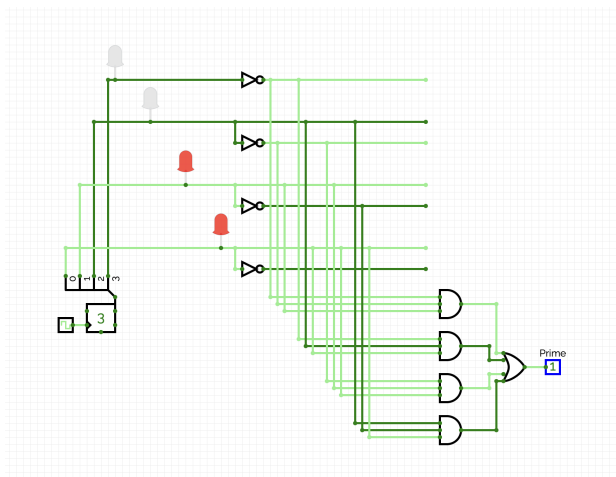
The Number 1 is neither prime nor composite which is why the output is False (0)

### Number 2:



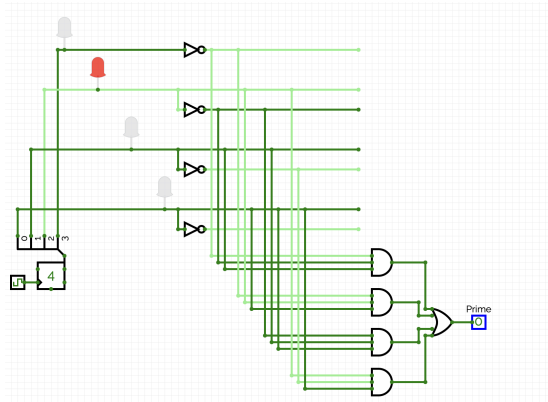
The Number 2 is prime which is why the output is True (1)

### Number 3:



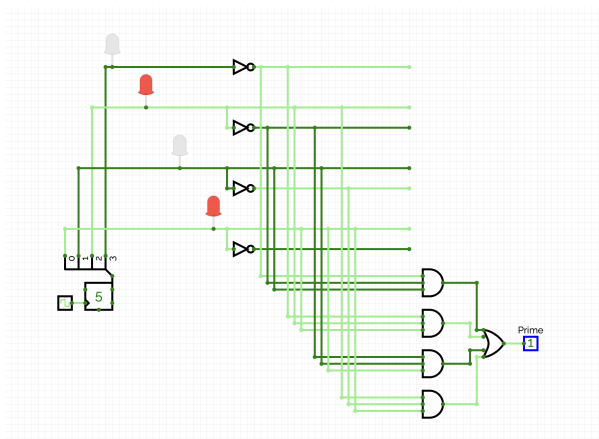
The Number 3 is prime which is why the output is True (1)

### Number 4:



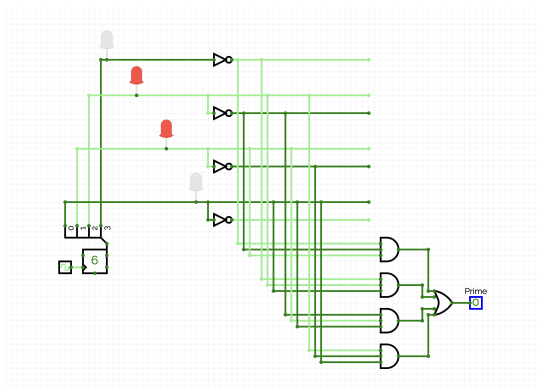
The Number 4 is Composite which is why the output is False (0)

**Number 5:**



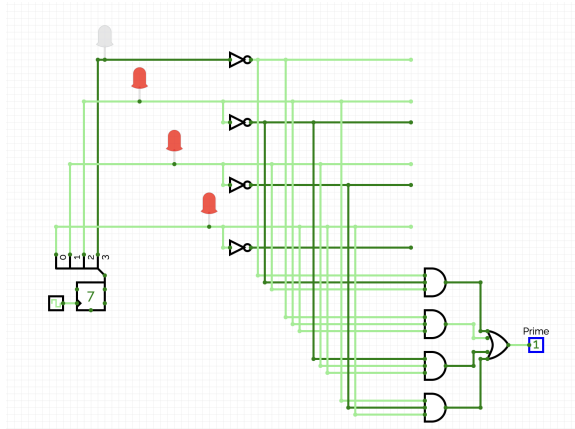
The Number 5 is Prime which is why the output is True (1)

**Number 6:**



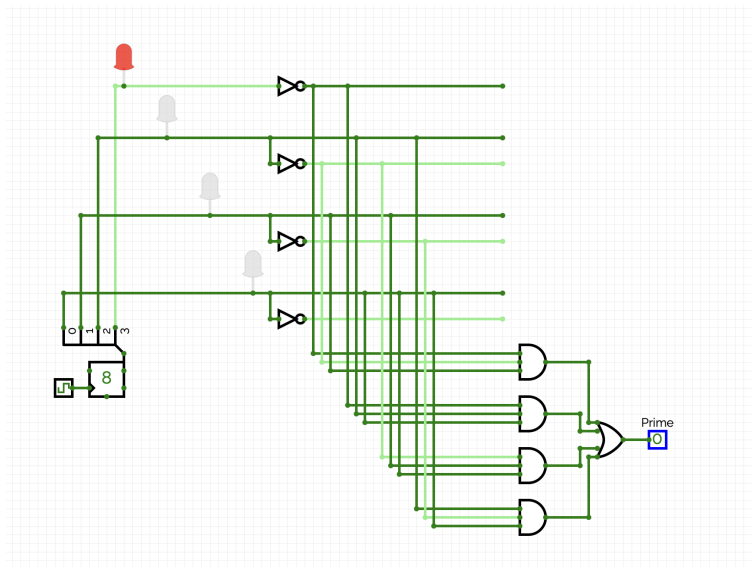
The Number 6 is Composite which is why the output is False(0)

**Number 7:**



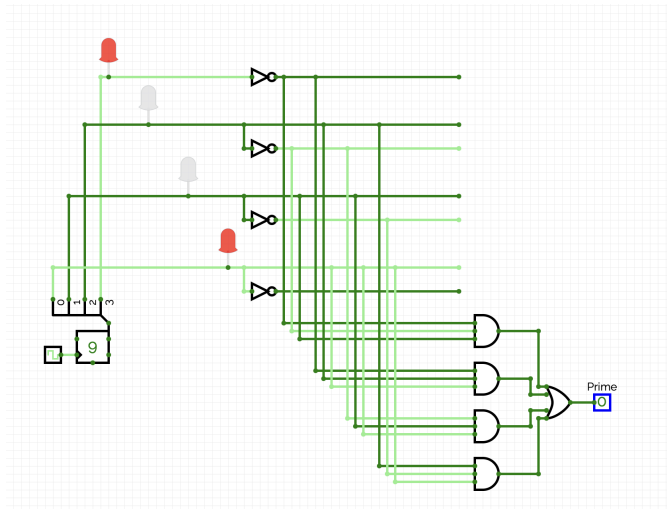
The Number 7 is Prime which is why the output is True (1)

### Number 8:



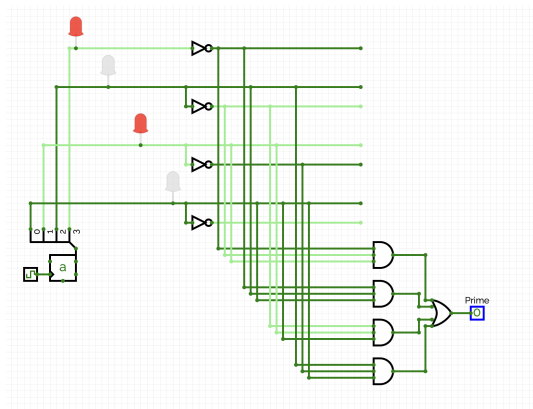
The Number 8 is Composite which is why the output is False(0)

### Number 9:



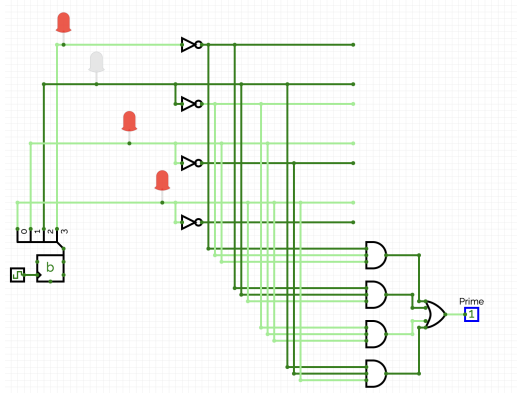
The Number 9 is Composite which is why the output is False(0)

### Number 10:



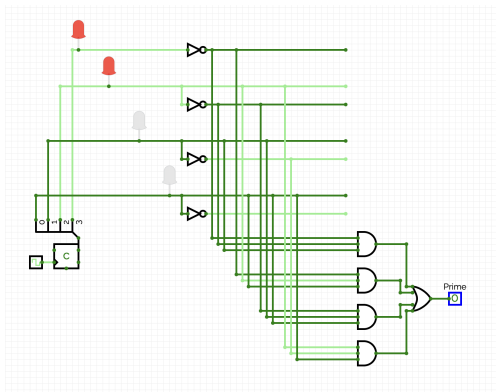
The Number 10 which is represented by a is Composite which is why the output is False(0)

### Number 11:



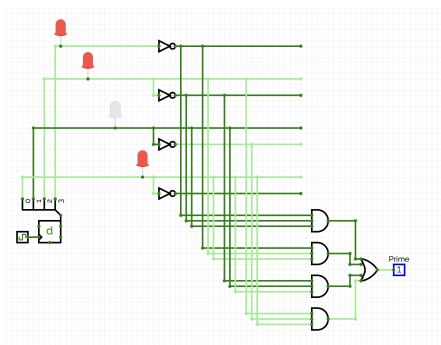
The Number 11 which is represented by b is Prime which is why the output is True (1)

### Number 12:



The Number 12 which is represented by c is Composite which is why the output is False(0)

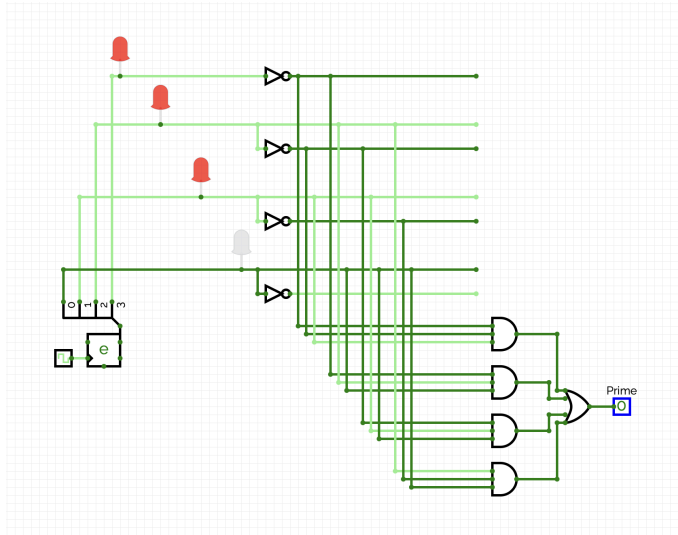
### Number 13:



The Number 13 which is represented by d is Prime which is why the output is True (1)

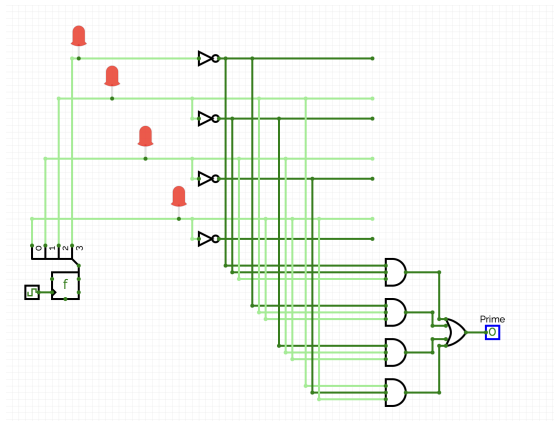
### Number 14:





The Number 14 which is represented by e is Composite which is why the output is False(0)

### Number 15:



The Number 15 which is represented by f is Composite which is why the output is False(0)

### Conclusion:

In this lab, we effectively created and tested a digital circuit to distinguish prime numbers between 0 and 15. And when we described prime numbers as numbers that have dividers only 1 and the number itself, resulted in a logic circuit which gives 1 for prime numbers and 0 for other numbers. Testing confirmed that the circuit returns 1 in case if the number is prime, 0 otherwise for such numbers as 2, 3, 5, 7, 11, 13 and so on. Many of these concepts proved especially practical in a combinational logic design and application of number theory in digital circuits.